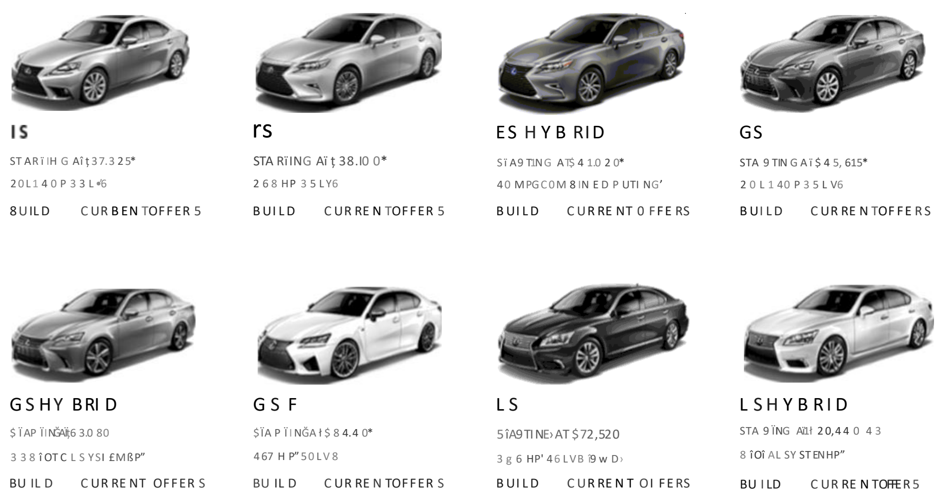
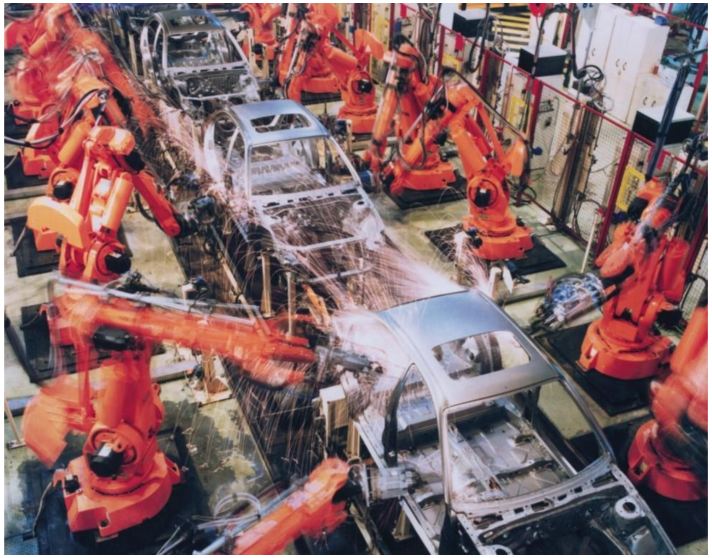
USABILITY

What happens outside (Customer?) and What happened inside (Engineering?)

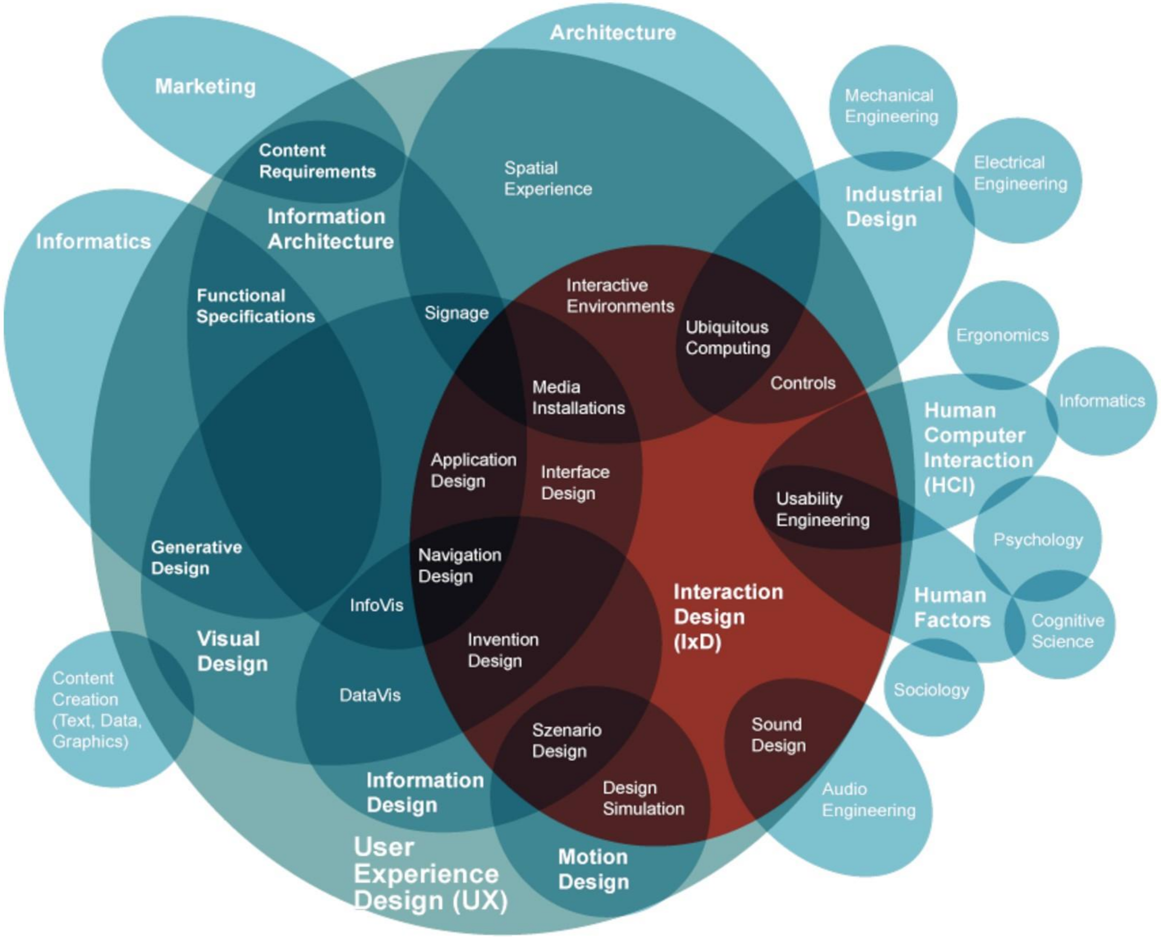




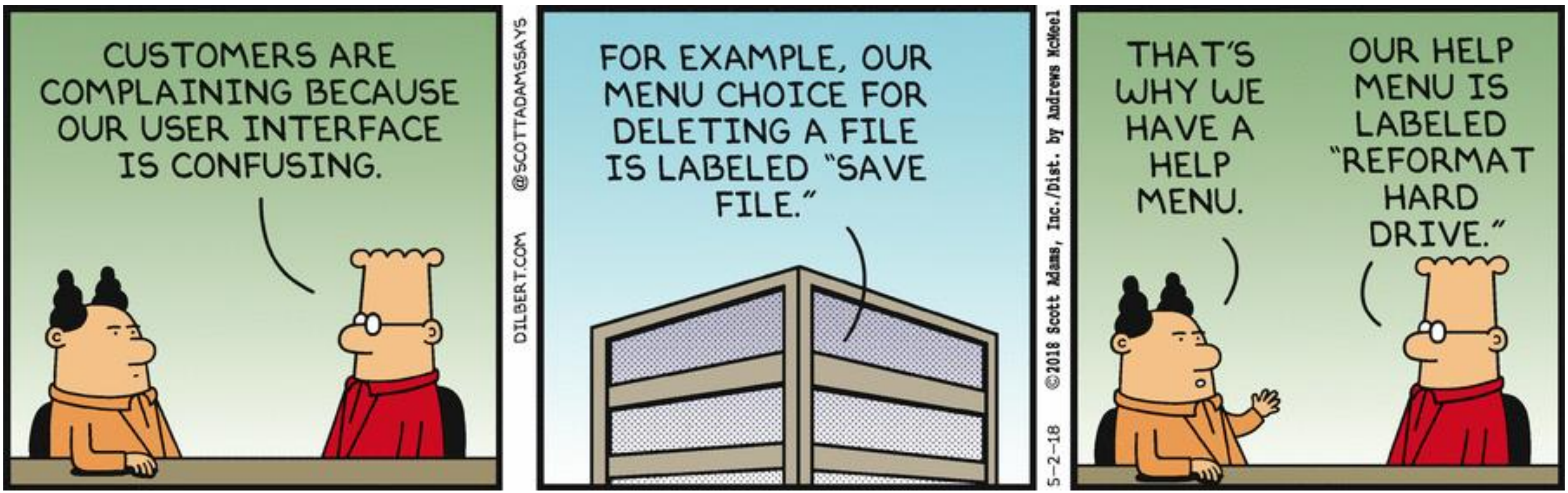
Different Design Aspects

- Architecture design
  - Division into subsystems and components
    - How these will be connected
    - How they will interact
    - Their interfaces
- Class Design
  - Various features of classes
- User Interface design
- Algorithm design
- Protocol design
  - Design of communication protocol

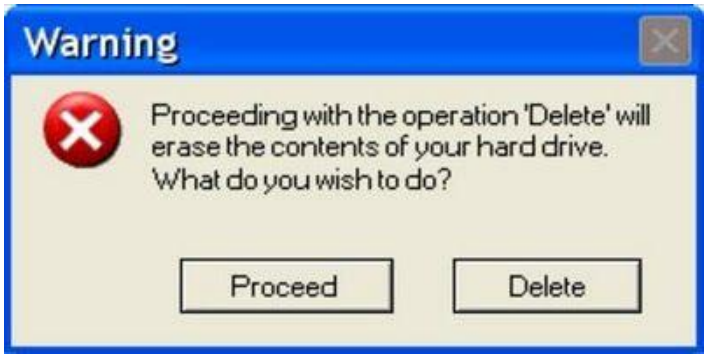
User Experience Design

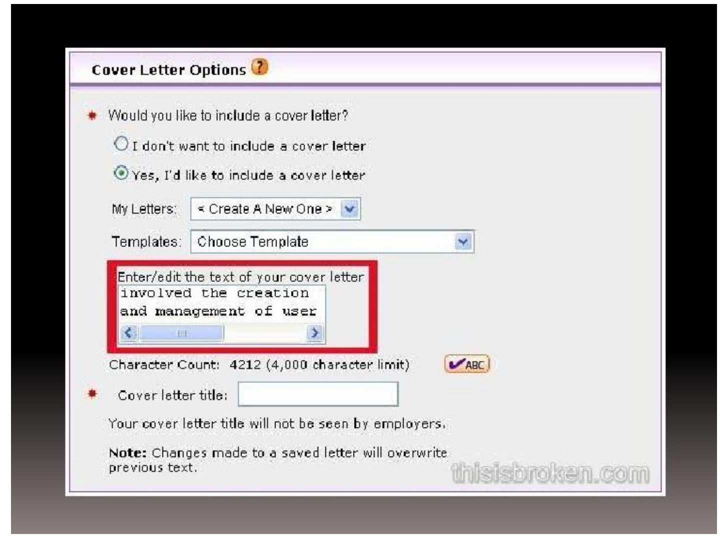


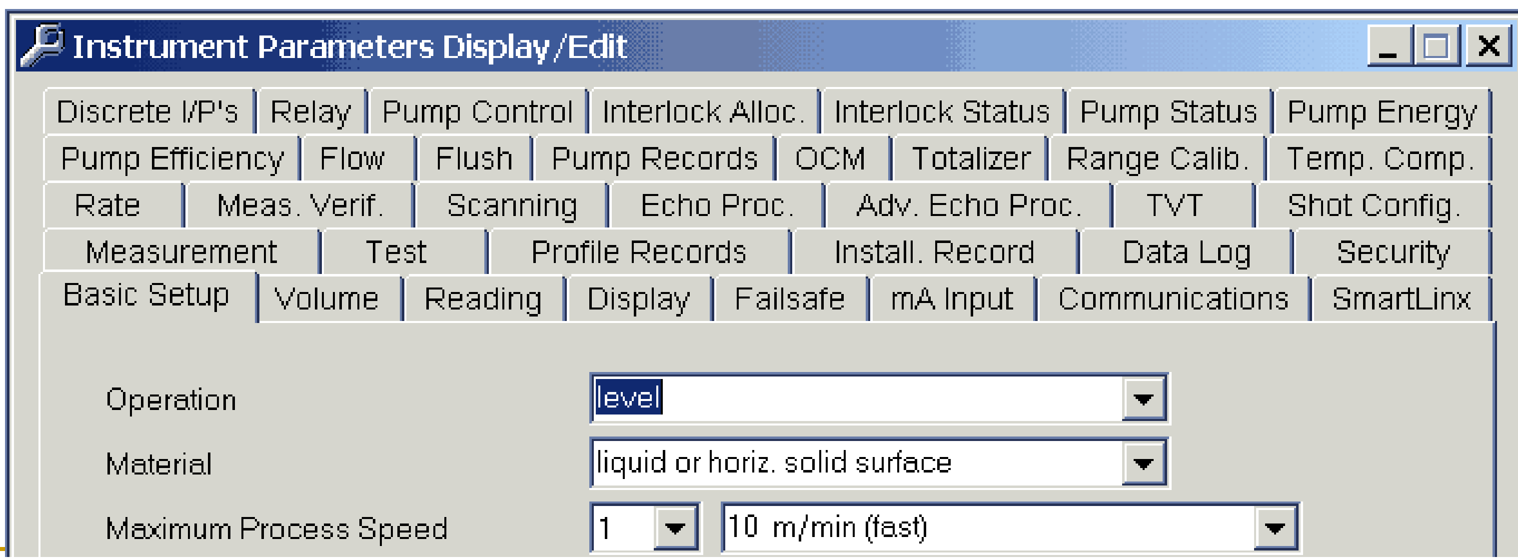
UI Failures


CUSTOMERS ARE COMPLAINING BECAUSE OUR USER INTERFACE IS CONFUSING.
DILBERT.COM @SCOTTADAMSSAYS
FOR EXAMPLE, OUR MENU CHOICE FOR DELETING A FILE IS LABELED "SAVE FILE."
©2018 Scott Adams, Inc./Dist. by Andrews McMeel 5-2-18
THAT'S WHY WE HAVE A HELP MENU.
OUR HELP MENU IS LABELED "REFORMAT HARD DRIVE."

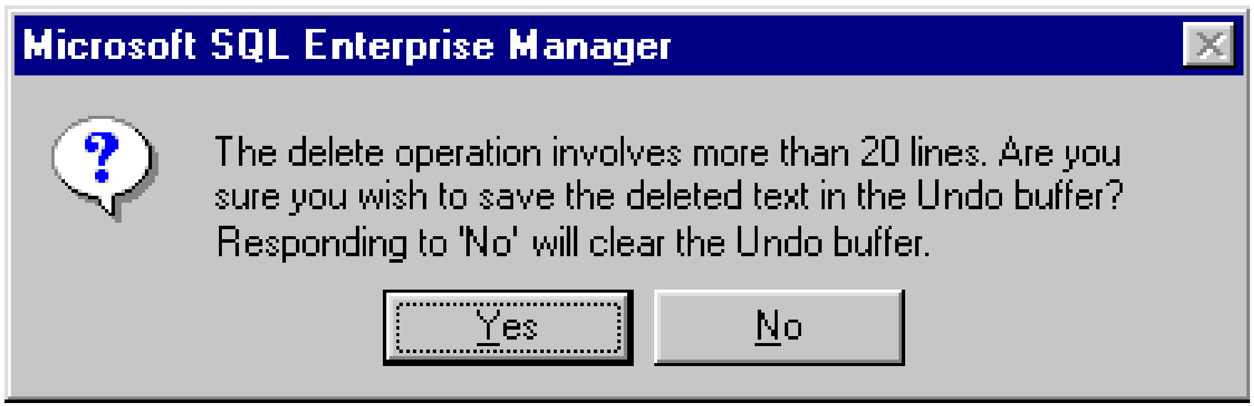
UI failures contd…

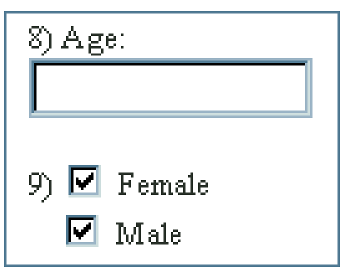


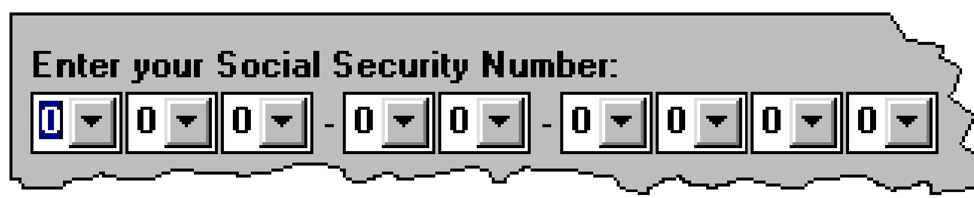




UI failures contd…





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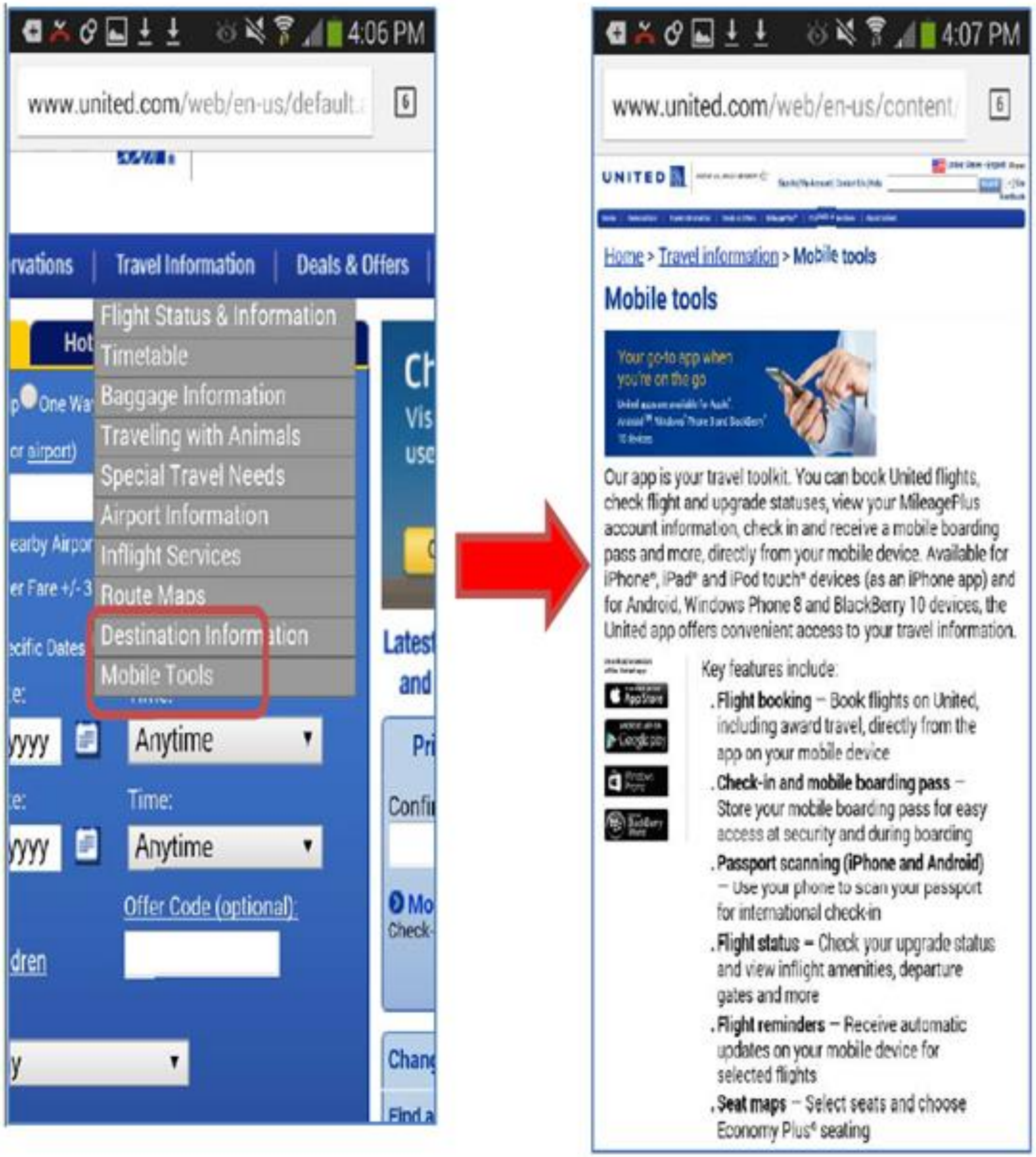
espn.go.com
THIS IS YOUR GAME
King's Reign Continues
THIS IS YOUR GAME
OTL: Will Committee Get It Right?


AT&T M-Cell 10:15 AM
espn.go.com
Pros Called Him "Straight Arrow"
Learn his Swing Secrets Right Here!
SquareToSquareMethod.com
SPORTS
PNC
King's Reign Continues
Expecting someone else? LeBron James takes the NBA's best talents back to Cleveland.
Predictions: Preseason awards
MORE
PREV
1/5
NEXT
NFL



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Travel Information
Deals & Offers
Flight Status & Information
Timetable
Baggage Information
Traveling with Animals
Special Travel Needs
Airport Information
Inflight Services
Route Maps
Destination Information
Mobile Tools
Anytime
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Mobile tools
Your go-to app when you're on the go
Our app is your travel toolkit. You can book United flights, check flight and upgrade statuses, view your MileagePlus account information, check in and receive a mobile boarding pass and more, directly from your mobile device. Available for iPhone®, iPad® and iPod touch® devices (as an iPhone app) and for Android, Windows Phone 8 and BlackBerry 10 devices, the United app offers convenient access to your travel information.
Key features include:
Flight booking – Book flights on United, including award travel, directly from the app on your mobile device
Check-in and mobile boarding pass – Store your mobile boarding pass for easy access at security and during boarding
Passport scanning (iPhone and Android) – Use your phone to scan your passport for international check-in
Flight status – Check your upgrade status and view inflight amenities, departure gates and more
Flight reminders – Receive automatic updates on your mobile device for selected flights
Seat maps – Select seats and choose Economy Plus® seating

Usability

- Broadly the above examples can be looked at from
  - Functional Perspective
  - User Perspective

Usability

ISO (ISO/IEC 9126-1:2001) defines usability as

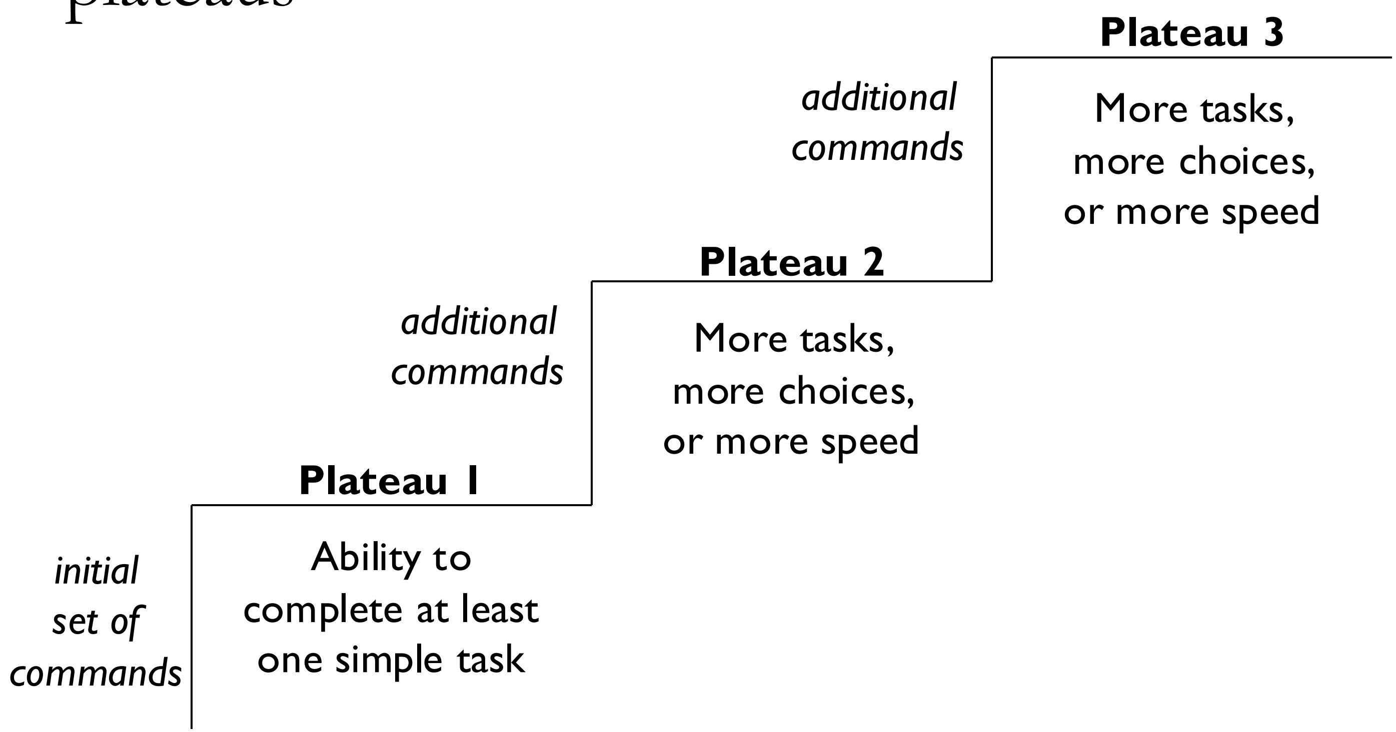
"The extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use"

Attributes for Evaluating Usability

i. Learnability
ii. Efficiency
iii. Memorability
iv. Errors
v. Satisfaction

1. Time to Learn (Learnability)

- How long it takes to learn how to use an interface
- With complicated interfaces, learning happens in "plateaus"



2. Speed of Performance (Efficiency)

- Speed of user interface, **NOT** software
- Number of characters to type, buttons to press, mouse-clicks, mouse movements, …
- Speed of performance often directly conflicts with time to learn
  - That is, faster systems are often harder to learn
  - Command lines vs. GUIs

3. Rate of User Errors

- A UI needs to be designed in such a way that user mistakes are less likely
- Affected by factors such as :
  - Consistency
  - Instructions
  - Logical arrangement of screens
- Importance depends on criticality of software

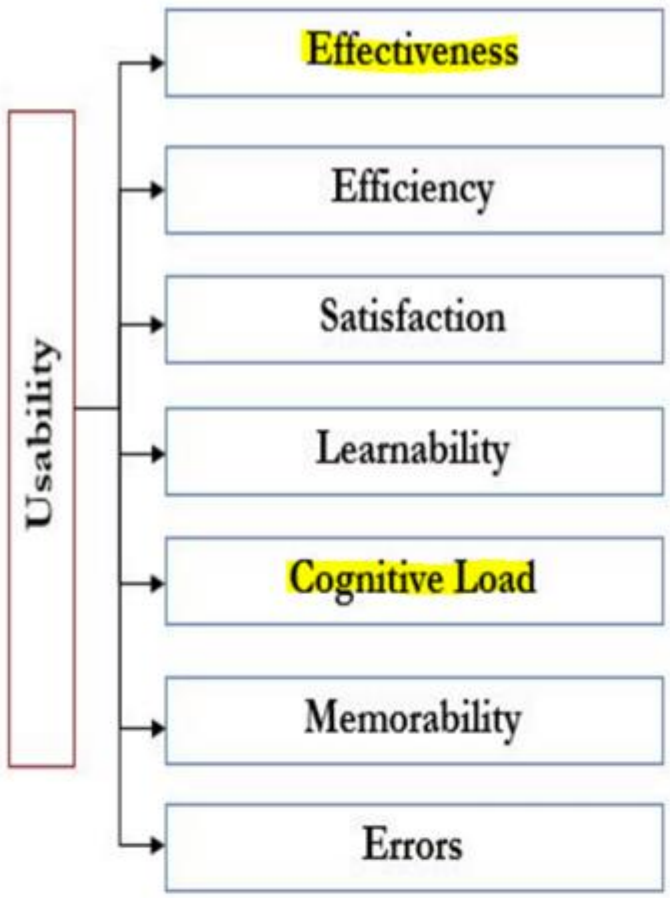
4. Retention of Skills (Memorability)

- We quickly forget how to use some user interfaces, but remember others for life
  - Calculus vs. algebra
  - Airplanes vs. bicycles
- Affected by how closely the syntax of the operations match our understanding
- If learning is very fast, retention may be less important

5. Subjective Satisfaction

- How comfortable the users are with the software
- Previous criteria are very analytical, objective, and measurable
- Subjective satisfaction captures other issues that are more specific to individual taste and background
  - A little harder to measure

People At the Centre of Mobile Application Development (PACMAD)



Design of User Interfaces

- Inside-out design :

  Develop a system, then add the interface

- Outside-in design :

  Design the interface, then build the system to support it

When decisions are made, either the developer must conform to the user, or the user must conform to the developer.

Traditional CS is entirely inside out

DZone Mobile Zone

Over a million developers have joined

EFCARDZ GUIDES ZONES JOBS | Agile AI Big Data Cloud Database DevOps Integration IoT Java Microservice

11 Reasons Mobile Apps Fail

The mobile industry is rapidly growing to an endless scope. But increased competition also leads to increased chances of failure.

by Lauren Gilmore · Jul. 05, 18 · Mobile Zone · Opinion

User Issues

There's nothing more frustrating than a mobile app that c
- or no performance - is the number one reason why users

6. Bad User Experience

Offering a compelling user experience is crucial for the suc
available features are key, user interface design is a major
difficult to use and causes performance issues.

Solution: Designate enough attention to UX/UI best practice

testbytes

Making Quality a Habit

Top 8 Reasons for App Failure and How to Avoid Them

Tuesday August 7, 2018

App Testing

5. Poor User Interface

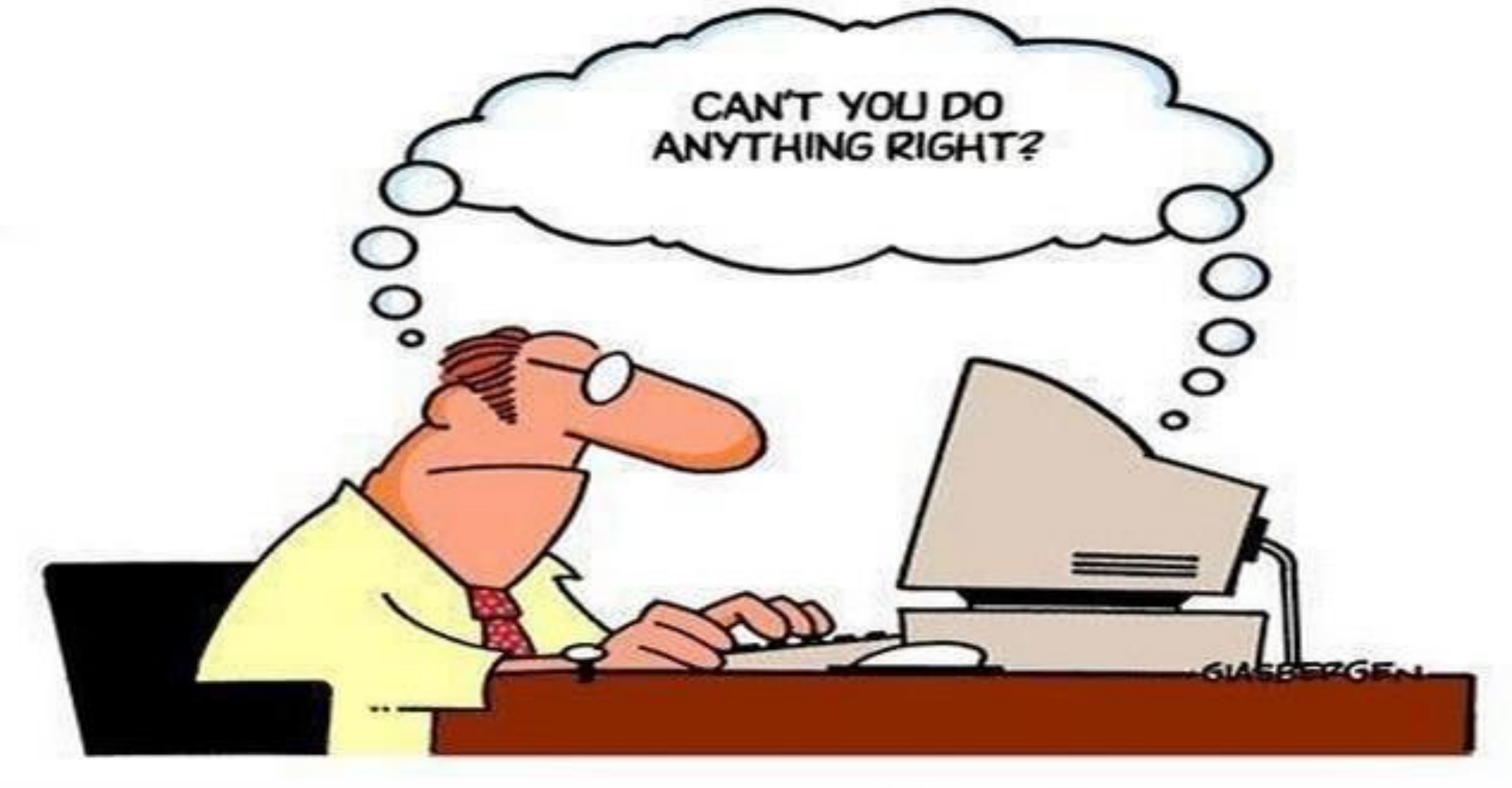
Generally, poor user interface means a poor design. Today, most apps are rejected by users ultimately leading to their failure due to poor user experience. If the performance of an app is unsatisfying to the users, they will uninstall it in no time. If users are unable to perform basic functions seamlessly, then it will be very hard for developers to sell their product. There are various reasons how an app can give poor user interface such as:

Why is Designing UIs Hard ?

- Good UIs take time to design
  - Many programmers are lazy
- Designing a good UI requires thinking like the user instead of an engineer
  - Engineers often think they are users
- Different users want different things

Engineers love features

They want to do everything the technology allows!


CAN'T YOU DO
ANYTHING RIGHT?
GLASBERGEN

Why is Designing UIs Hard ?

- Designers are usually experts
  - They view all functions as having equal weight
- Marketing want UIs designed for beginners
  - They sell to beginners
  - Many have only novice semantic knowledge
- Most users are in between—intermediates !

Don’t weld on training wheels

Understand the Users

- Work experience
- Computer experience
- Age
- Education
- Reading skills
- Language skills
- Work environment
- Task frequency
- … many more possibilities

It is important to know who the user is

Distribution of Users

Beginners

What does the application do?

What doesn't it do?

How do I print?

Where do I start?

Intermediates

Forgot how

Where is it?

Remind me …

Can I undo?

More features?

Experts

Automate?

Shortcuts?

Can I change?

Customize?

Keyboard shortcut?

Dangerous?

UI Patterns

http://ui-patterns.com/patterns

Shneiderman’s Golden Usability Principles

1. Strive for consistency
2. Design usable and discoverable shortcuts
3. Provide appropriate feedback
4. Yield closure
5. Provide appropriate error handling
6. Allow users to reverse (undo) all actions
7. Support internal locus of control
8. Reduce the Short Term Memory (STM) load

Principle (1/8) - Consistency

MacOS menu bar remains consistent! 1980s to 2019 !

Various aspects of UI Must be consistent ! Changes break Consistency!!

Some examples from Interaction Design Foundation

Principle (2/8) – Shortcuts

Design usable and discoverable shortcuts

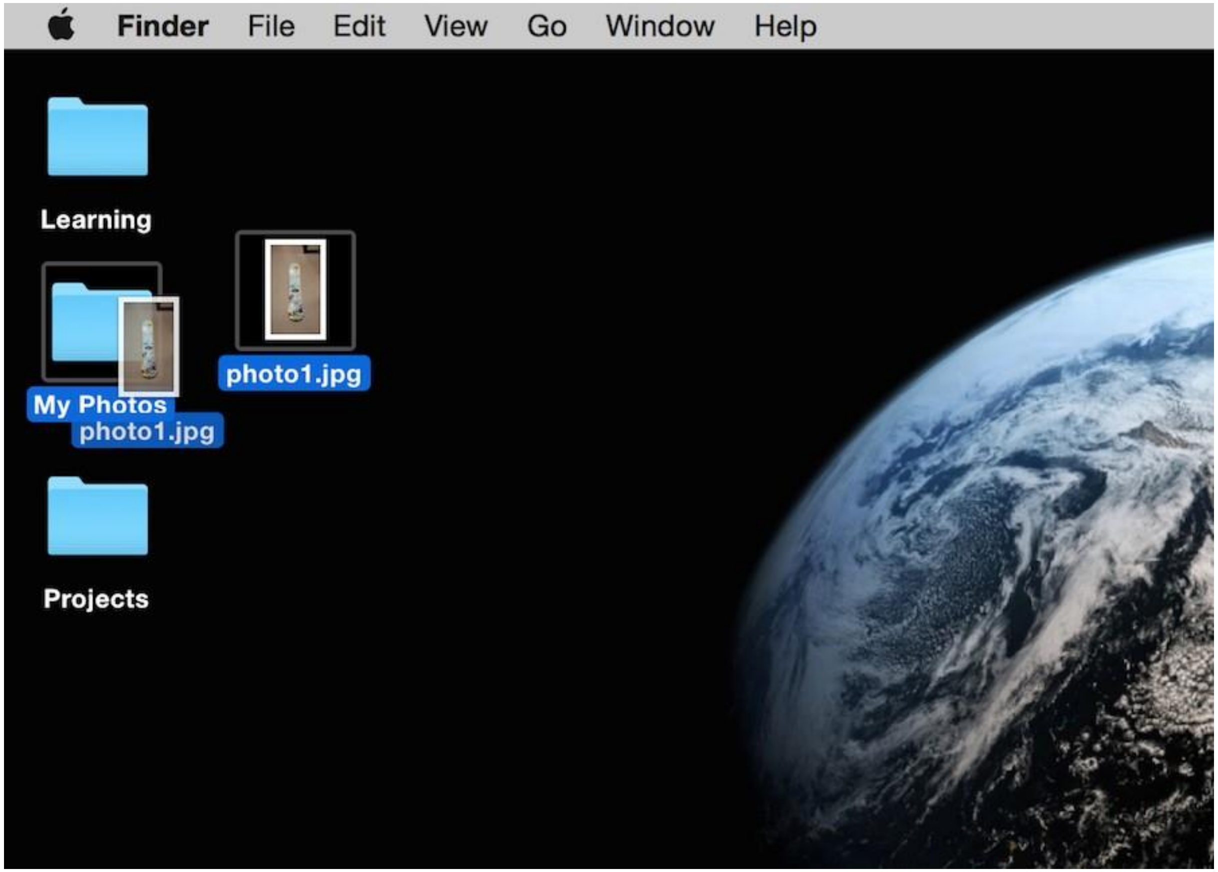
- Users must be able to find them
- Users must be able to remember them

  - Copy and Paste (Command-C and Command-V)

  - Screenshots (Command-Shift-3)

Some examples from Interaction Design Foundation

Principle (3/8) - Feedback

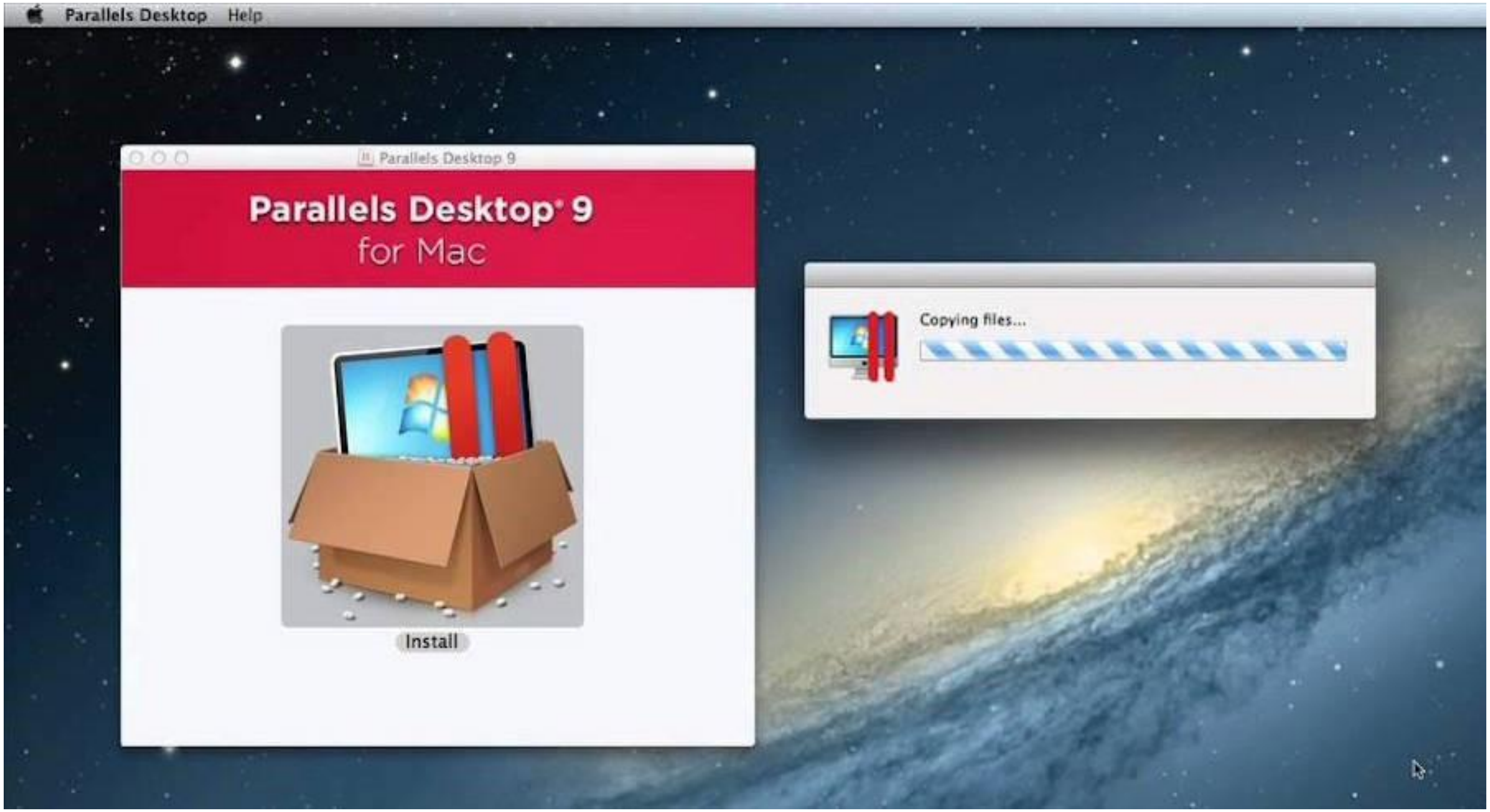


Folder is shown to be physically moving during drag and drop

Have a clearly defined end-point in the interaction !

Some examples from Interaction Design Foundation

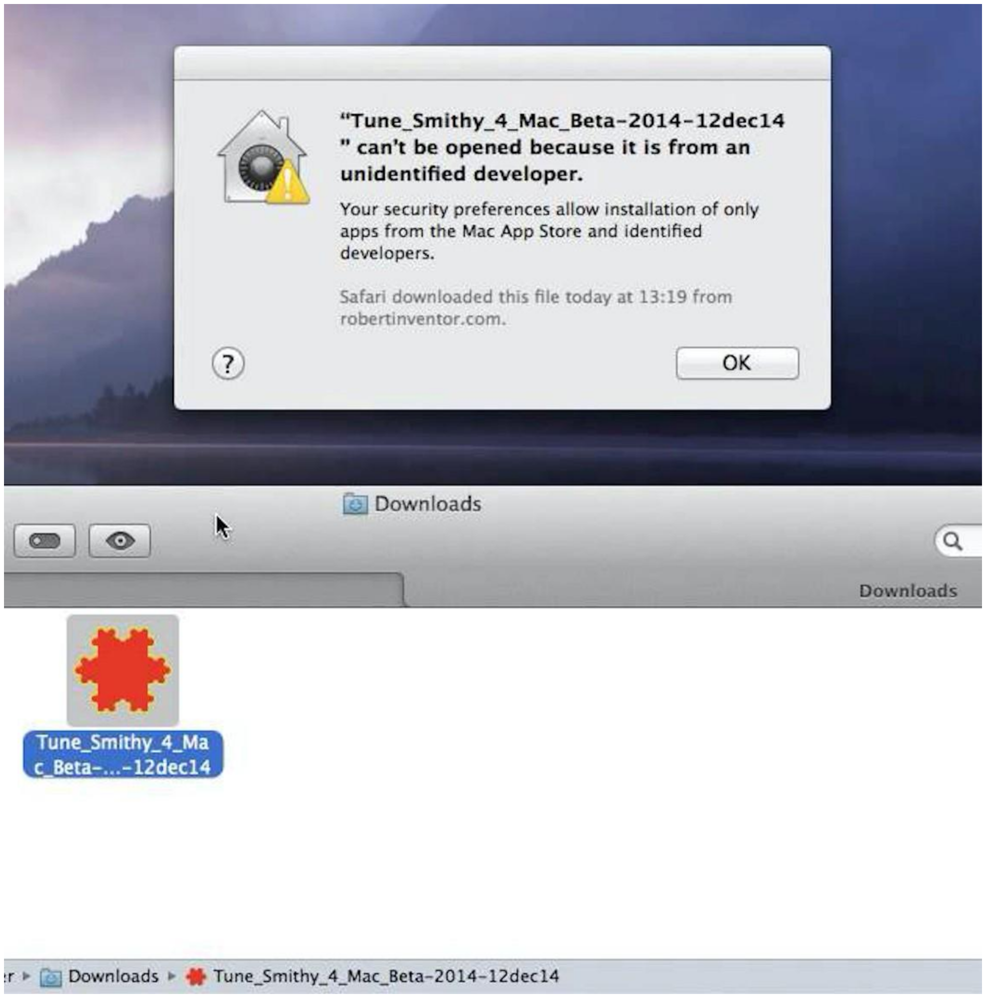
Principle (4/8) – Yield closure



Appropriate feedback must be given (For errors as well as others) !

Some examples from Interaction Design Foundation

Principle (5/8) – Error handling



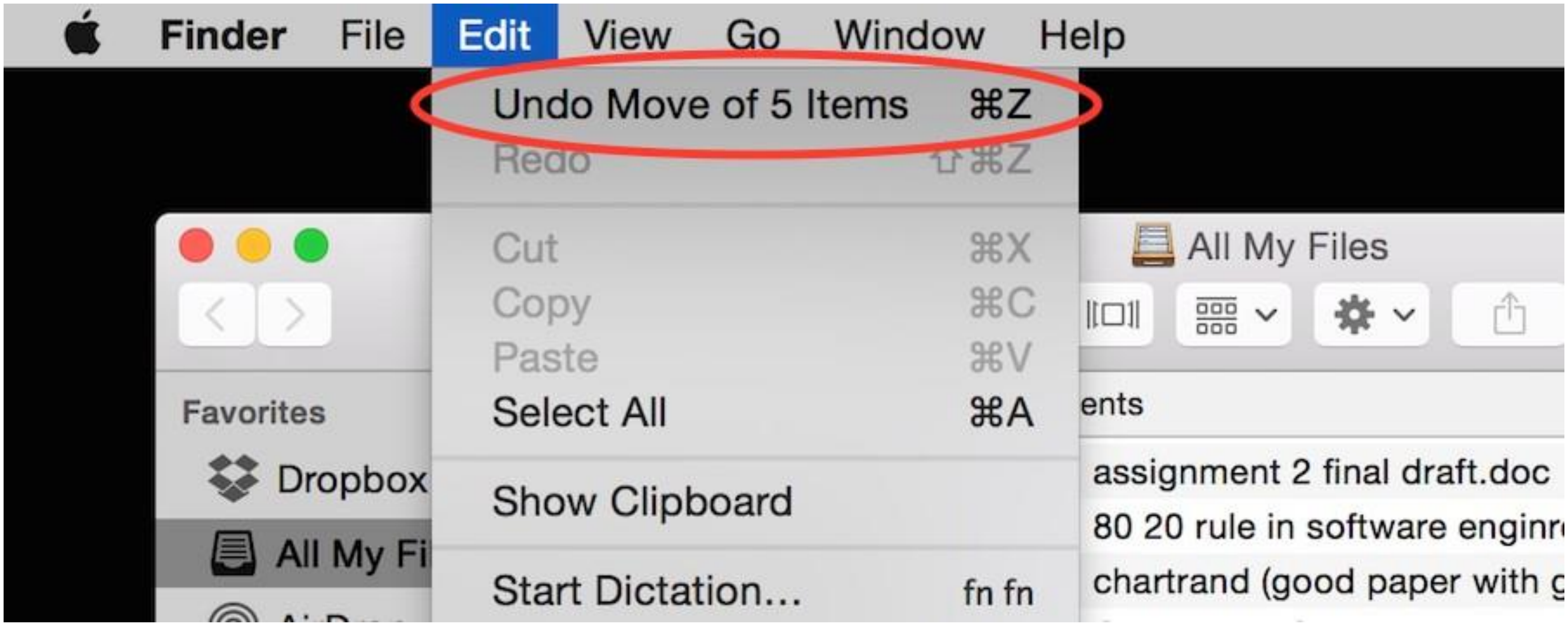
- Clearly tell users what was wrong
- Don't scare the users. Use appropriate language
- Don't just give Error handles or Error codes
  - "404 error" does not really convey much
- Only make users redo the part that was wrong

Some examples from Interaction Design Foundation

Principle (5/8) – Error handling

Some examples from Interaction Design Foundation

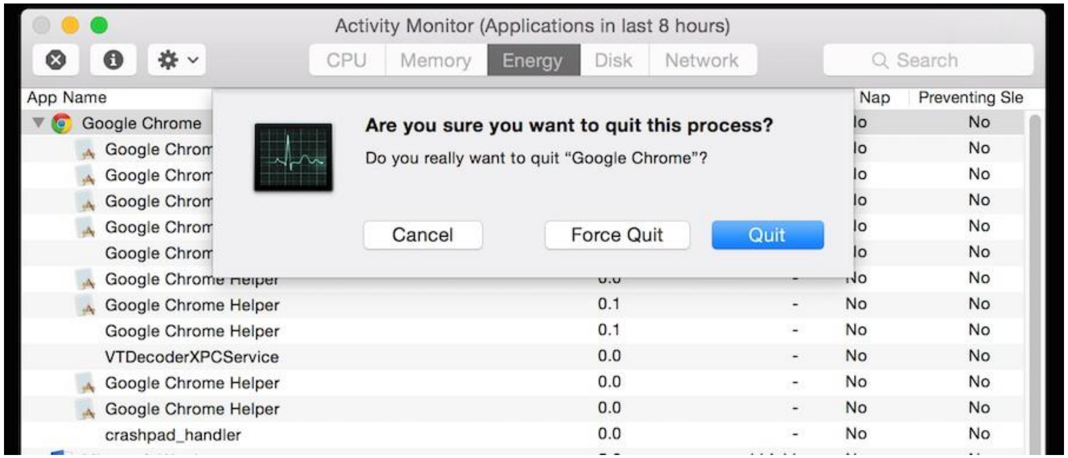
Principle (6/8) - UNDO

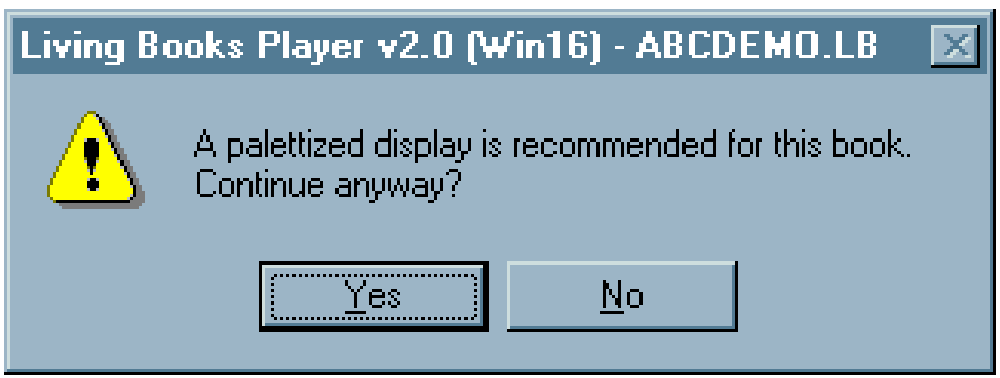


Actions must be reversible ! If some actions can't be undone, use "hesitation"

Some examples from Interaction Design Foundation

Principle (7/8) – Internal locus of control





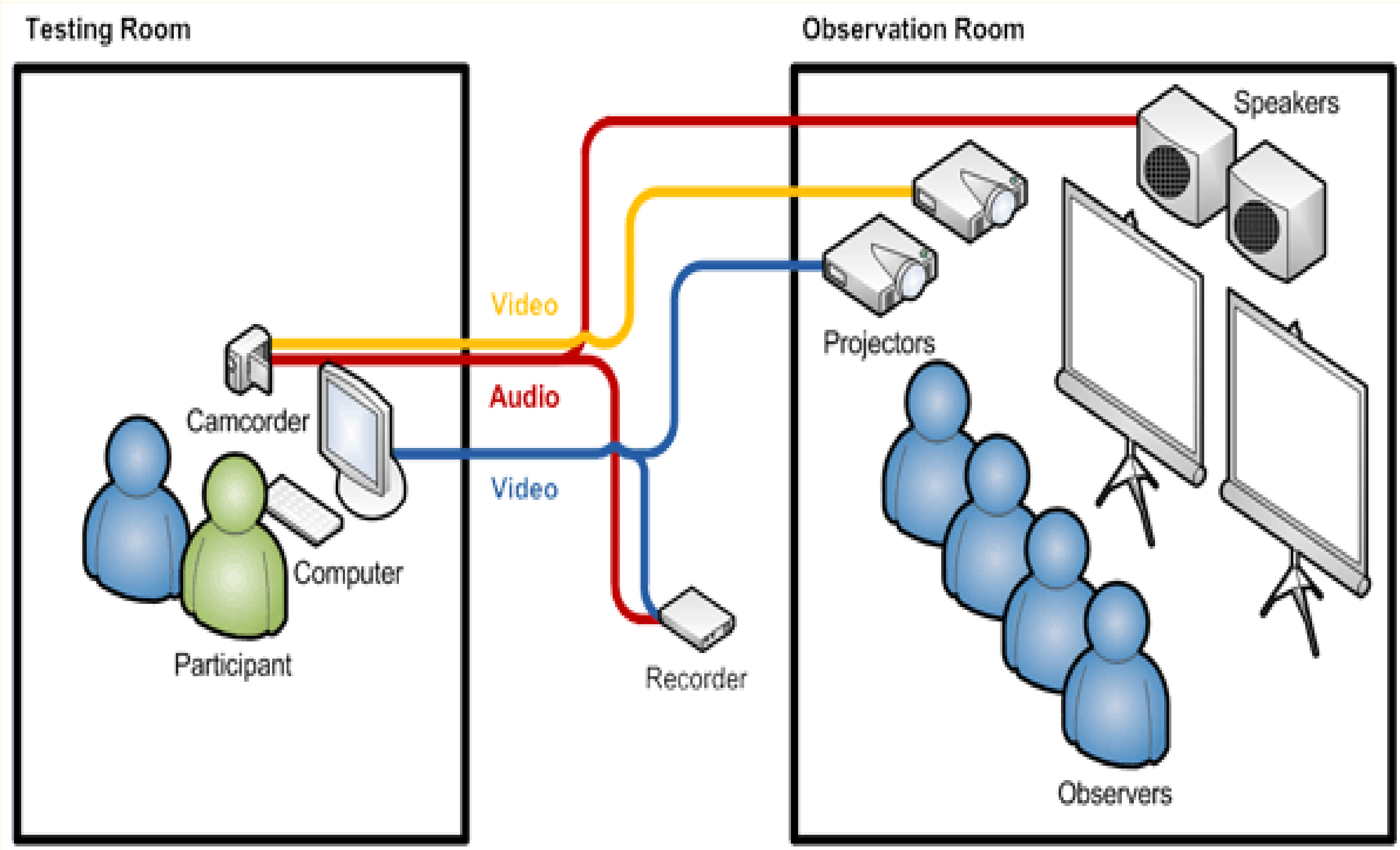
- Inexperienced users may be intimidated when the software makes decisions
- Experienced users want to control the flow
- Put the users in-charge (to the extent possible)
  - Hide technical terms
  - Interaction must be customizable (Personalization)

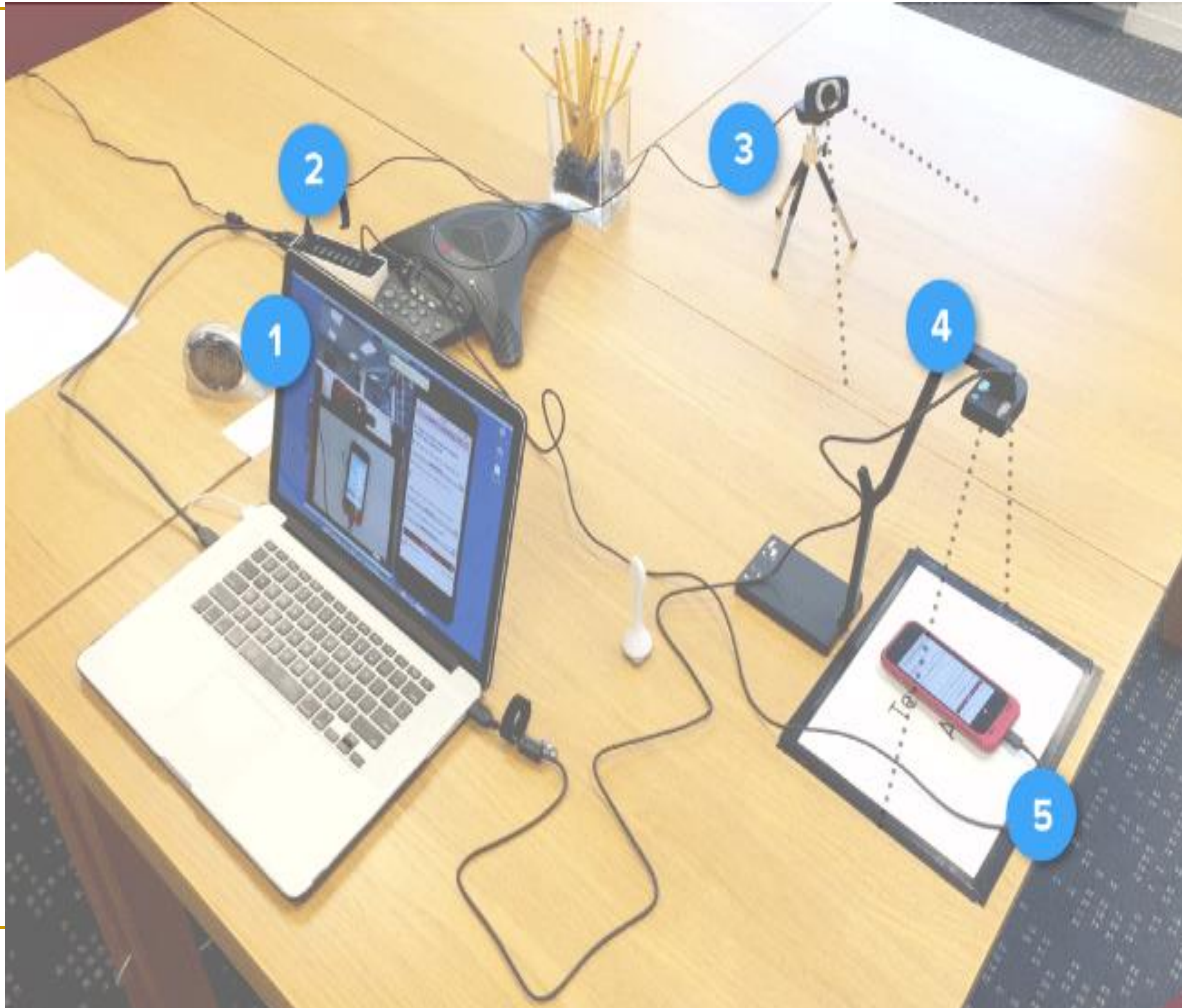
Some examples from Interaction Design Foundation

Principle (8/8) – Short term Memory (STM)

- **Remember 7 +/- 2**
- **Apple has only 4 apps in the main menu**
- **Organizing things hierarchically help reduce STM**

Usability Labs

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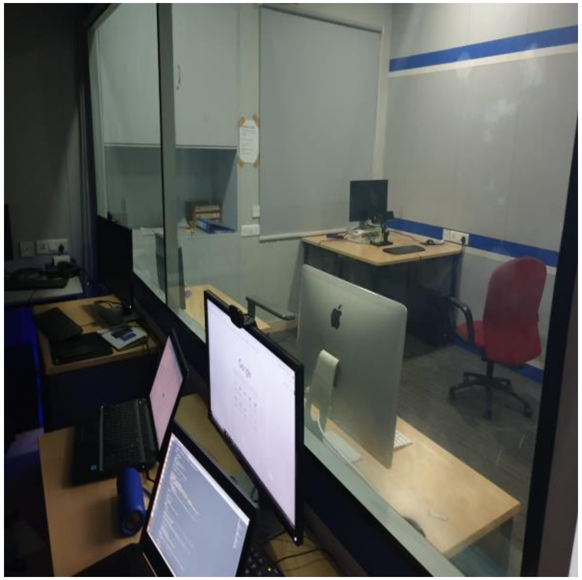
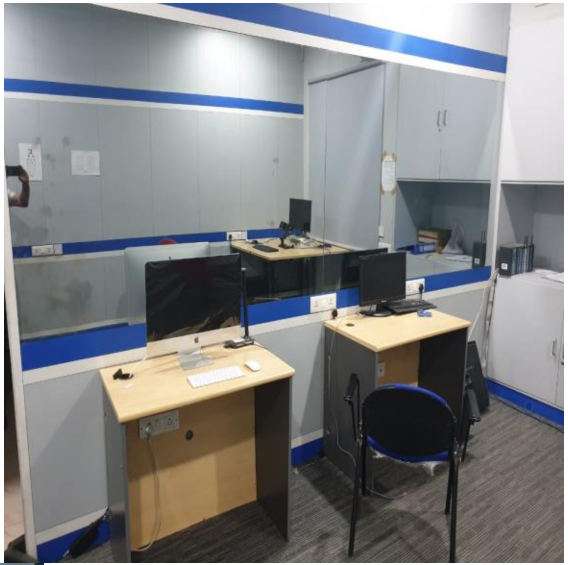
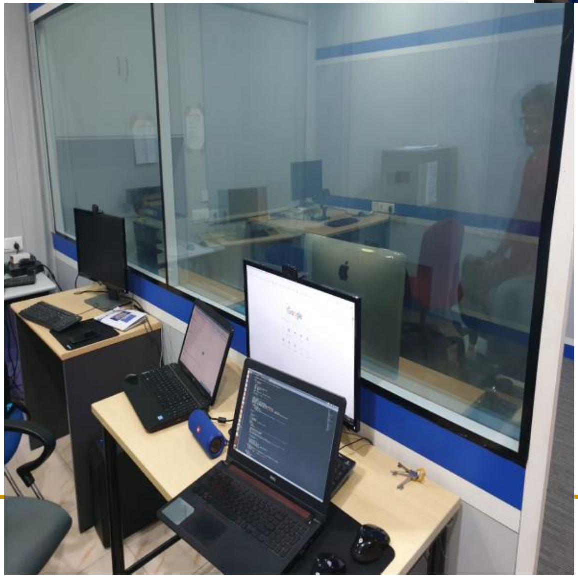
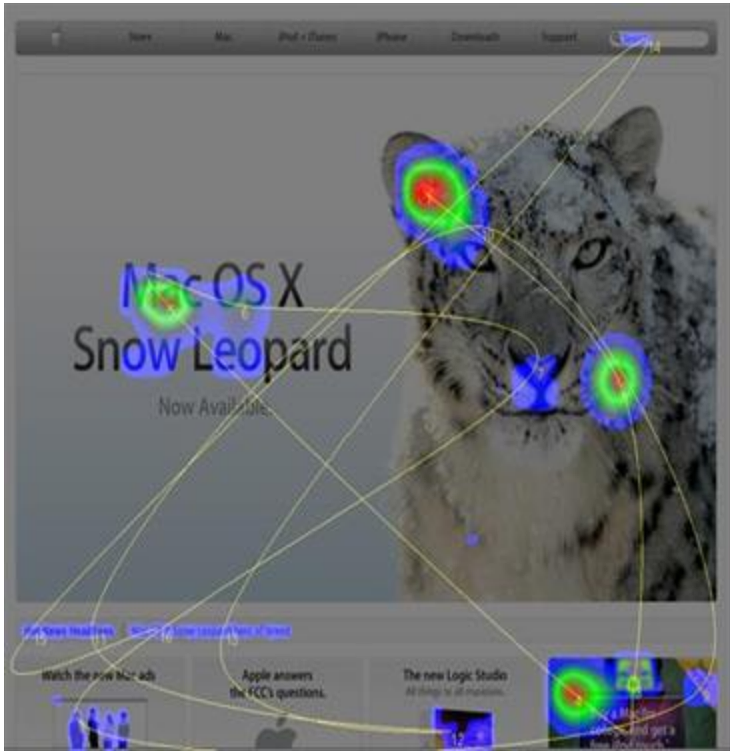
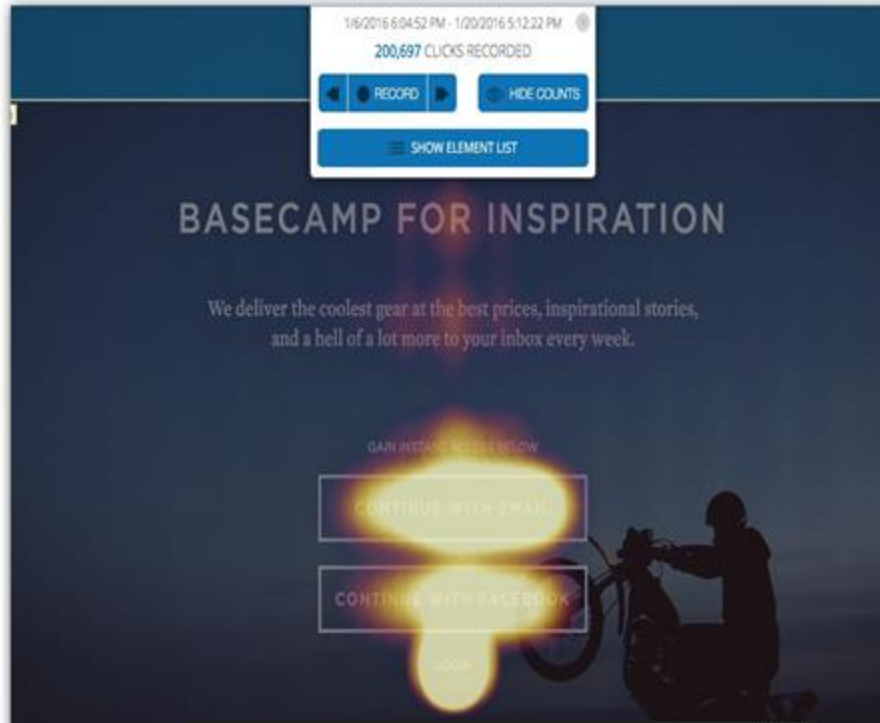
Virtual Human Interaction Lab (@ SERC, IIITH)








Snow Leopard
Apple answers
the FCC's questions.
The new Logic Studio


200,697 CLICKS RECORDED
RECORD
HIDE COUNTS
SHOW ELEMENT LIST
BASECAMP FOR INSPIRATION
We deliver the coolest gear at the best prices, inspirational stories,
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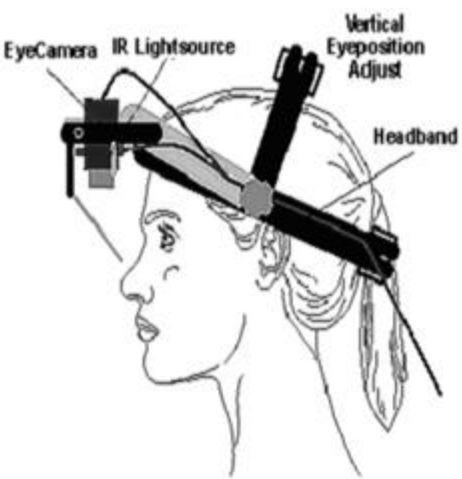
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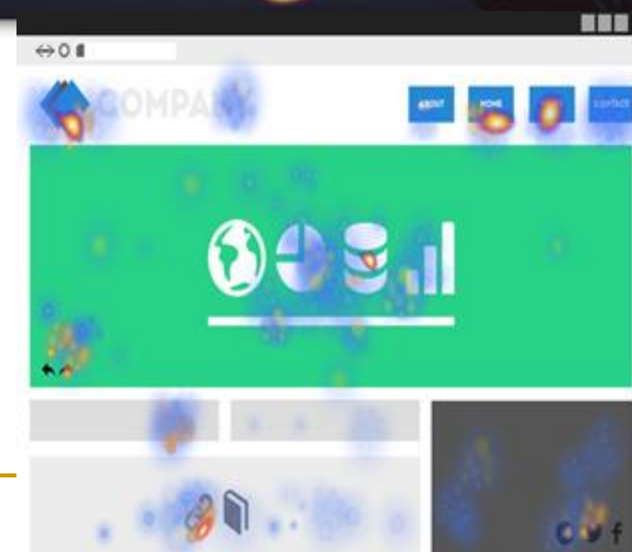

Usability Guidelines

Usability Guidelines are statements by which to determine an action of usefulness. A Usability guideline aims to streamline particular processes according to a set routine or sound practice. By definition, a guideline is never mandatory. Guidelines are not binding and are not enforced.

Usability guideline examples

- In a form, telephone number entry is restricted to a dial-pad (i.e. numbers only)
- User can select to reveal or hide password as they type, during signup or sign-in (e.g. by toggling a 'reveal' or 'hide' control)
- If app requests sign-up, the user can choose to continue as a guest
- First-time user has multiple sign-up options (e.g.,username/password and social sign-up facebook.com, google, twitter)